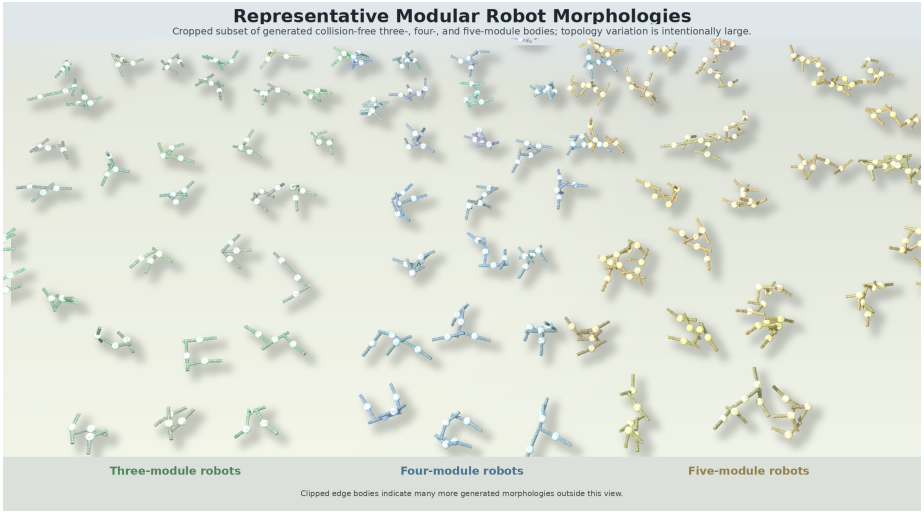
Learning cross-topology control

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figureRepresentative modular robot morphologies. The benchmark includes collision-free three-, four-, and five-module bodies with large topology variation, so evaluation tests transfer across unseen attachment graphs rather than only size or link-parameter changes within one canonical body plan.

Abstract

Currently, whenever a novel robot is created, a novel policy must also be created to control it. Here we introduce a controller capable of transferring to out-of-distribution heterospecific robots with previously unseen topological structures. Our approach, the Model based CPG replanner, uses proprioceptive/action history to infer how a new body responds, then replans using a fixed library of CPG motion primitives. During training, rollouts on seen morphologies teach the model how different action primitives affect robot dynamics. When placed in a novel morphology, the controller performs a short warmup, uses the resulting sensorimotor history to estimate which primitives are likely to move that body well, selects one primitive, and revises that choice online. We evaluate this idea in a modular robot benchmark that realizes a wide diversity of novel body plans, instead of merely

enlarging or shrinking presupposed links within a canonical kinematic tree. This benchmark asks whether a controller can move beyond conspecific transfer, where prior work often remains within the same quadrupedal, bipedal, or otherwise fixed body form. Across three-, four-, and five-module robot families, we train on 100 seen morphologies and test on 1000 held-out morphologies with unseen attachment graphs. In matched 25 s MuJoCo evaluations, the Model based CPG replanner obtains the largest mean net displacement on all three families, suggesting a path toward reconfigurable robots that can form new structures for new tasks without rebuilding control from scratch.

Locomotion, universal control, cross-embodiment, modular robots.

1 Introduction

Why Cross-Embodiment Locomotion Matters. Robot design is becoming increasingly open-ended: bodies can be resized, reconfigured, or assembled for different tasks, but locomotion control is still often tied to a specific embodiment. When each new body requires its own policy, the controller becomes a bottleneck on morphology exploration. This is especially limiting for reconfigurable robots, where the value of the hardware comes from creating many possible bodies rather than perfecting one fixed form.

Why Modular Robots Are the Right Testbed. Instead of merely enlarging or shrinking presupposed links within a canonical kinematic tree, we use a modular robot kit to realize a wide diversity of novel body plans. The same physical modules can be reconnected into different topologies, so the hardware, sensing, and actuators remain fixed while the body plan changes. This lets us test whether a controller can adapt to new attachment graphs, rather than only interpolate over familiar shapes.

Beyond Conspecific Transfer. Prior work often stays within conspecific transfer: the controller moves among variants of the same quadrupedal, bipedal, or otherwise canonical body form. But there is no proof that these familiar body plans are optimal for every task. If a reconfigurable robot can shapeshift through a much larger space of structures, then the control problem becomes urgent: how do we quickly control each new structure well enough to explore the rest of the morphology space?

Why Topology Transfer Is Hard. Changing the attachment graph changes more than appearance. It changes which parts touch the ground, how forces travel through the body, and how each motor command affects motion. A controller that works on one modular assembly may therefore fail on another even though the modules, joints, and sensors are the same.

Limitation of Direct Policy Transfer. Many universal-control baselines directly predict low-level joint actions from history or morphology tokens. This asks one network to both infer the current body and synthesize stable motor commands at every step. In

a large topology space, this can make generalization brittle because the policy must extrapolate both body response and action structure at the same time.

Core Idea. We separate these two problems. Instead of predicting raw actions, our controller chooses among a fixed library of smooth Central Pattern Generator (CPG) action primitives. Learning then becomes an outcome-prediction problem: given recent sensorimotor history from the current body, which CPG primitive is most likely to produce useful displacement?

Benchmark and Claim. Each robot is assembled from repeated actuated modules, and each morphology is a graph whose nodes are modules and whose edges are physical dock-to-dock connections. Figure shows representative three-, four-, and five-module bodies from this topology-diverse benchmark. For each module-count family, we train and calibrate using 100 seen morphologies and evaluate on 1000 held-out morphologies with unseen attachment graphs. The paper's central claim is that seen-morphology rollouts can supervise a Model-Based CPG Replanner that selects and revises useful CPG actions for unseen modular bodies in real time.

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2 Methods

Pipeline Overview. This paper aims to train a controller that can choose useful locomotion primitives for a new modular body from a short history of how that body responds. The controller should not need to identify the morphology explicitly or run many candidate actions in simulation at deployment time. To achieve this, the method has three training/deployment stages: collect rollout chunks from seen morphologies, label each chunk with the outcome of many candidate CPG actions, and train a CPG replanner scorer with a privileged teacher and deployable RMA student that predicts the best CPG for a new morphology from recent proprioceptive/action history. Figure 1 summarizes the full training and deployment pipeline.

Seen-Morphology Rollout Chunks. We first run locomotion rollouts on the 100 seen morphologies. These rollouts provide realistic histories: joint states, proprioceptive signals, previous actions, and contact situations that the bodies naturally enter during locomotion. Rather than treating a full rollout as one training example, we cut it into short chunks, so each chunk becomes a local sensorimotor context for the current morphology.

Why Chunks Matter. Two identical CPG actions can have different effects depending on whether the robot is stalled, turning, tilted, or already moving. Chunk-level supervision therefore teaches the scorer to condition on recent history, not only on the candidate CPG parameters.

Fixed CPG Library. We define a fixed bank of 4096 CPG primitives. Each primitive is a smooth periodic joint-position command with per-joint amplitude, phase, and offset parameters plus one shared frequency. The same bank is used for supervision, selection, and replanning.

Parallel CPG Outcome Labeling. For each rollout chunk, we restore the same simulator state and execute many candidate CPG primitives from that identical starting point. The displacement after the rollout horizon becomes the outcome label for that candidate. This creates a counterfactual table: from the same body state and history, which CPG would have moved the robot best?

Training Signal. The label for candidate c_k at chunk s from morphology i is the horizontal displacement:

$$r_{i,s,k} = \left\| x^{xy}_{i,s,k}(t+H) - x^{xy}_{i,s}(t) \right\|_2 . \tag{1}$$

This turns locomotion learning into CPG outcome prediction rather than direct action imitation. After the parallel offline evaluation of many CPGs from many chunk states, the deployable model learns from these labels so it does not need to repeat that search online.

CPG Replanner Scorer. The scorer has two versions: a privileged teacher used only during training and a deployable RMA student used at test time. The teacher receives the candidate CPG features c_k and a privileged body/state summary $p_{i,s}$ from the restored rollout chunk. It encodes them into a candidate latent z_c and a privileged body latent z_p, then predicts the CPG outcome with a prior term plus a body-conditioned interaction term:

$$z_c = g_c(c_k), \qquad z_p = g_p(p_{i,s}), \tag{2}$$

$$\hat{r}^T_{i,s,k} = q(z_c) + \beta\, u([z_c, z_p, z_c \odot z_p]) . \tag{3}$$

Here q is the candidate prior head, u is the interaction head, and $\odot$ denotes elementwise multiplication. The prior estimates whether a CPG is generally useful, while the interaction term estimates whether it is useful for this particular body state.

The deployable RMA scorer cannot use privileged simulator summaries, morphology ID, or hidden state. It builds a history latent from recent proprioceptive/action history:

$$z_h = e_\theta(h_{i,s}). \tag{4}$$

The history encoder first summarizes each module's recent tokens and actions, then pools those module embeddings into one body-response latent. Deployment uses the same frozen candidate encoder and score heads as the teacher, but substitutes z_h for the privileged latent z_p:

$$\hat{r}^D_{i,s,k} = q(z_c) + \beta\, u([z_c, z_h, z_c \odot z_h]) . \tag{5}$$

Training aligns the deployable scorer with the teacher by matching teacher scores, matching the teacher privileged latent, and ranking CPG candidates according to the

expert rollout displacement labels $r_{i,s,k}$. Figure 1 shows this distillation structure together with online deployment on an unseen morphology.

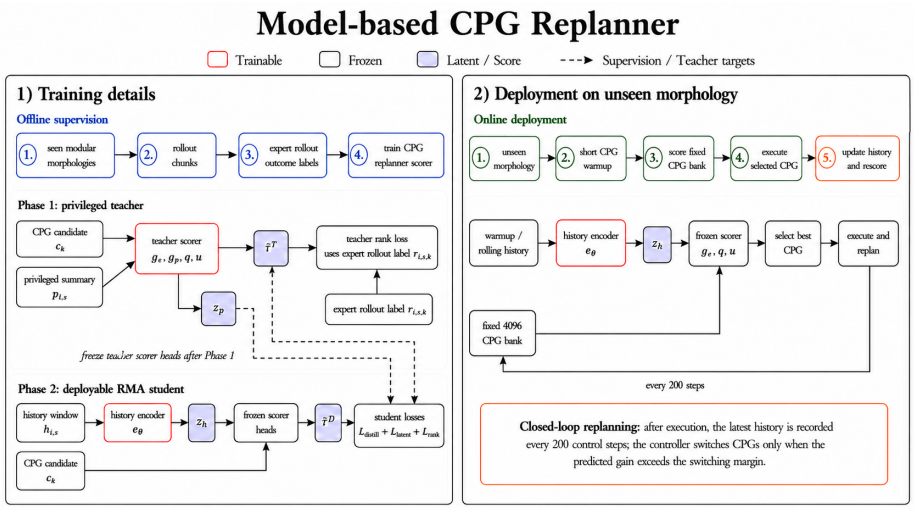


Figure 1: Model-based CPG Replanner pipeline. Panel 1 summarizes offline training: seen modular morphologies are rolled out, cut into chunks, labeled with expert rollout outcome labels $r_{i,s,k}$, and used to train the CPG replanner scorer. The privileged teacher learns from candidate CPGs and privileged summaries, then the deployable RMA student learns a history latent that can replace the privileged latent while the teacher scorer heads are frozen. Panel 2 shows deployment on an unseen morphology: warmup and rolling history are encoded, the fixed 4096-CPG bank is scored, the best CPG is executed, and the history is updated for replanning every 200 control steps.

What the Model Learns. Training on many chunks across many seen morphologies teaches the scorer how different action sequences affect different body dynamics. If a new morphology twists, stalls, or accelerates during warmup, the history encoder can use that response to change the CPG ranking.

Real-Time Deployment. At deployment, the robot first executes a short random CPG warmup. This is not for optimization; it is for measurement. The resulting sensorimotor history reveals how the unseen body responds to actuation, and the deployable scorer uses that history to choose the CPG predicted to produce the largest displacement.

Closed-Loop Replanning. The selected CPG is not fixed forever. Every 200 control steps, the controller updates its history and rescores candidates. It switches only when the predicted improvement over the current primitive exceeds a calibrated switching

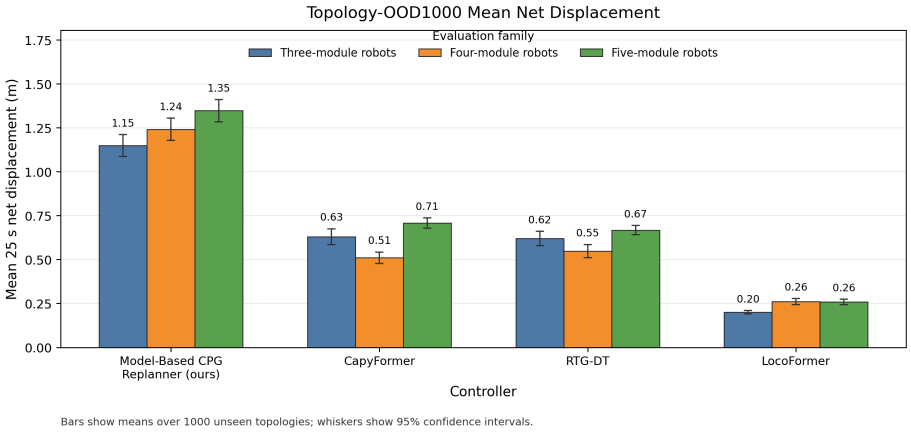


Figure 2: Topology-OOD1000 mean net displacement. Bars report mean 25 s net displacement over 1000 unseen topologies for each module-count family; whiskers show 95% confidence intervals. The Model-Based CPG Replanner obtains the largest mean displacement for three-, four-, and five-module robots.

cost. This lets the controller recover from a poor first choice while avoiding noisy changes away from a gait that is already working.

3 Results

Main Evaluation Question. The main experiment asks whether a scorer trained on 100 seen morphologies can control 1000 unseen morphologies from the same module-count family. We evaluate three-, four-, and five-module families under matched MuJoCo conditions.

Baseline Comparison. We compare against universal neural controllers that directly predict actions, including RTG-DT, CapyFormer, and LocoFormer. These baselines test whether CPG outcome prediction and replanning provide a better transfer mechanism than direct low-level action prediction.

Cross-Morphology Transfer. Across the three-, four-, and five-module held-out morphology sets, the Model-Based CPG Replanner obtains the largest mean net displacement under the matched evaluation protocol (Figure 2). The mean 25 s net displacement is 1.15 m, 1.24 m, and 1.35 m on the three-, four-, and five-module families, respectively. This supports the claim that learned CPG outcome prediction transfers across unseen topologies.

Replanning Ablation. We compare the full replanner with a one-shot CPG selector and a periodic random CPG selector. This separates the value of the learned scorer

Controller	Trials	Mean net displacement ↑ (m)
Model-Based CPG Replanner	3000	**1.780 ± 0.072 (1.111)**
One-shot CPG selector	3000	1.709 ± 0.083 (0.836)
Periodic random CPG selector	3000	1.119 ± 0.031 (0.913)

Table 1: Replanning ablation on 3000 unseen modular morphologies.

Entries are mean 50 s net displacement in meters with 95% confidence intervals; medians are shown in parentheses. The table pools 1000 three-module, 1000 four-module, and 1000 five-module held-out morphologies. The paired gain over one-shot selection is $+0.071$ m, with one-sided paired t-test $p = 0.0055$.

from the value of merely changing CPGs over time. The full replanner improves over one-shot selection in a separate 50 s ablation protocol (Table 1), showing that new sensorimotor evidence during the rollout can improve gait choice.

Efficiency of Learned Prediction. The controller does not need to run 4096 simulations for an unseen morphology at test time. The offline counterfactual labels train a scorer that can predict useful CPGs from recent history, making deployment much cheaper than online action search.